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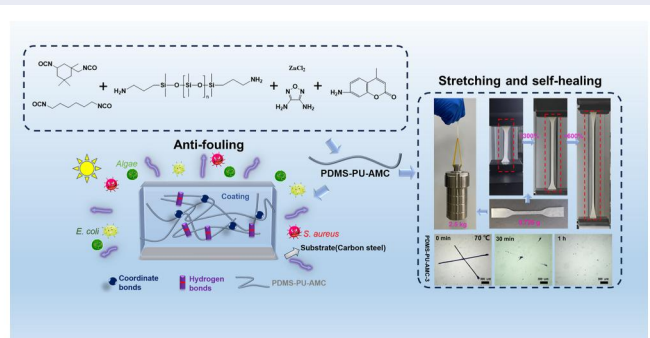
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ABSTRACT

Global demand for coatings with ship anti-corrosion and anti-fouling function is increasing every year. In this study, isocyanate terminated PDMS has been prepared by the reaction of isocyanates with amino groups. 7-amino-4-methylcoumarin (AMC) and 3,4-diaminofurazan (DAF) were chemically grafted onto silicone-based polyurea molecules to obtain anti-fouling coating (PDMS-PU-AMC-x) which exhibits excellent mechanical and anti-fouling properties. The dynamic multi-urea hydrogen bonds within the polymer chains and zinc ions coordination with the urea groups ensure the mechanical strength of the coating. The mechanical properties of the coating can be adjusted by the different molar ratios of two isocyanates for hexamethylene diisocyanate (HDI) and isophorone diisocyanate (IPDI). The silicone and two isocyanates act as soft and hard segments which provide flexibility and mechanical strength, respectively. The tensile strength of the PDMA-PU-AMC-3 achieve 11.8 MPa accompanied with tensile deformation 1073.1% when the ratio of IPDI to HDI is 1:1. The dynamic hydrogen bonds and coordination bonds enable the self-healing efficiency of PDMS-PU-AMC-3 up to 65% after 6 h at 70 °C. The synergistic antibacterial effect of DAF and AMC supplies excellent antimicrobial and antifouling capabilities. This work overcomes the low strength of traditional silicone-based self-healing materials and addresses a new strategy for marine coatings.

GRAPHICAL ABSTRACT



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Silicone; two isocyanates; high mechanical strength; synergistic effect; marine antifouling coating

1. Introduction

Marine biofouling means that fouling organisms grow and accumulate on the surface of underwater equipment, which exerts a serious negative impact on people's production activities.^[1] Fouling organisms adhering to the ship surface increase the navigation resistance and fuel consumption.^[2] The rate of metal corrosion can be accelerated on account of adhering fouling organisms.^[3,4] The marine biofouling and corrosion issues have severely affected the development of the marine economy.^[5] With the increasing consciousness of environmental protection in various countries, there exists

an urgent demand to develop a friendly coating to marine environment, antifouling and resistant to corrosion.^[6]

Numerous types of antifouling coatings were researched to achieve highly effective antifouling, such as traditional antifouling coatings,^[7] and silicone-based fouling-release coatings.^[8] Traditional antifouling coatings based on tributyltin (TBT) are widely used due to their excellent antifouling effects, but TBT-based coatings have a high teratogenicity rate to marine organisms and accumulate in marine organisms, causing harm to marine organisms.^[9] Some antifouling coatings also contain biocides and heavy metal ions, which are toxic to

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marine fouling organisms or other non-target marine organisms.^[10,11] The above antifouling coatings seriously damage the marine ecological environment, so they are gradually banned.^[12] Hydrolyzed self-polishing antifouling coatings are capable of releasing toxic substances and antifouling agent, which pollutes marine environment. Polydimethylsiloxane (PDMS) is frequently applied to make materials with specific functions such as low surface free energy and good flexibility.^[13,14] However, silicone-based fouling-release coatings often show a weak adhesion to the substrate material, the coatings can easily divestiture from the substrate surface by exterior shear forces, and its anti-fouling ability is not ideal under static conditions.^[15,16] Silicon-based fouling-release antifouling coatings are sensitive to exterior damage by virtue of their poor mechanical properties.^[17] These shortcomings limit its application.

To solve the aforementioned problem, some effective efforts have been focused to improve the mechanical properties, adhesion strength, antifouling effect and service life of the coatings. The mechanical strength and adhesion strength of the material can be raised by increasing the quantity of hydrogen bonds in the polymer system. Zhang et al. synthesized a viscous elastomer that can achieve fast, super-strong, and repeatable adhesion to a variety of surfaces through abundant hydrogen bonds in polymer system.^[18] However, this elastomer does not have the antifouling ability. Traditional polyurethane coatings can enhance antifouling effects by introducing amphiphiles,^[19] zwitterions,^[20] quaternary ammonium salts (QASs),^[21] and loading of antifoulants in polymer matrix.^[22] Sun et al. introduced tannic acid into a silicone-based polyurethane system, which significantly improved antimicrobial ability but deteriorated its mechanical properties. Self-healing ability is important for repairing material damage that can extend the material's service life.^[23] The intrinsic self-healing materials relying on dynamic exchange of covalent bonds were widely recognized, include hydrogen bonds,^[24] metal-ligand coordination,^[25] host-guest interactions,^[26] ionic interactions,^[27] π - π stacking,^[28] and disulfide bonds.^[29] In addition, the self-healing ability of the material is also related to the proportion of soft and hard segments in the polymer system and the fluidity of the molecular chains.^[30] The self-healing efficiency of materials gradually decreases with increase in the content of hard segments within the material. Better chain mobility leads to higher self-healing efficiency, but it also results in lower mechanical strength. Hence, it is difficult to achieve high mechanical strength and self-healing at the same time.^[31]

In the current research, a silicon-based polyurea material with excellent mechanical properties, good self-healing properties, anti-fouling, and anti-corrosion properties has been developed. The soft silicone section and two hard isocyanate segments ensure the flexibility and mechanical strength of the coatings. The self-healing properties are provided by the synergistic effect of hydrogen bonds and coordination bonds. To achieve high mechanical strength of the material while retaining its self-healing ability, the optimal ratio of two isocyanates with different structures has been explored.

The chain extender 3,4-diaminofurazan (DAF) and the antimicrobial monomer 7-amino-4-methylcoumarin(AMC) were introduced into the silicon-based polyurethane system by chemical grafting. More importantly, the urea bonds from the amino of DAF and isocyanates increase the number of hydrogen bonds in the polymer system. The high-density urea groups in the polymer network form hydrogen bonds on the surface of the substrate, which improve the adhesion strength to the substrate. The synergistic effect of DAF and AMC significantly enhances the antimicrobial and anti-corrosion capabilities of the coatings. This paper provides an approach for the development of a silicon-based coating with multiple functions.

2. Experimental

2.1. Materials

Aminopropyl terminated polydimethylsiloxane (APT-PDMS, $M_w = 1000$) was purchased by Hangzhou Silong Material Technology Co., Ltd. 3,4-diaminofurazan (DAF, 98%), 5-isocyanato-1-(isocyanatomethyl)-1,3,3-trimethylcyclohexane (IPDI, 98%) and hexamethylene diisocyanate (HDI, 98%) were bought from Shanghai Macklin Biochemical Co., Ltd. 7-amino-4-methylcoumarin(AMC, 98%) was bought from Bide Pharmatech., Ltd. Tetrahydrofuran (THF, A.R) and ethanol (A.R) were purchased from China National Medicines Corporation., Ltd. All monomers and reagents are used directly without purification.

2.2. The synthesis of isocyanate-terminated PDMS

The synthesis routes of the PDMS-PU-AMC-3 are shown in Figure 1 with three steps. First, 6 mmol IPDI and 6 mmol of HDI were dissolved in 20 mL THF. Then the solution was transferred to 250 mL three-neck flask equipped with nitrogen gas. 10 mmol APT-PDMS ($M_w = 1000$) was dissolved in 20 mL THF. The APT-PDMS solution was added in the above three-neck flask and the reaction temperature remained at 40 °C. The isocyanate-terminated PDMS was obtained after 4 h.

2.3. Synthesis of the PDMS-PU-AMC-x

DAF was added to the precursor (isocyanate-terminated PDMS) and the reaction was maintained at 40 °C about 3 h. Then, 0.171 mmol AMC was added into the three necked flask and the temperature was raised to 70 °C with another 3 h. Next, 0.33 mmol $ZnCl_2$ was put into the reaction system with stirring about 1 h. The final product was named PDMS-PU-AMC-x. The synthesis of the PDMS-PU-AMC-3 is shown in Figure 1. According to different ratios of IPDI:HDI (10:2, 8:4, 6:6, 4:8, 2:10), PDMS-PU-AMC-x ($x = 1, 2, 3, 4, 5$) were prepared successfully (Table S1). The obtained solution was cast on the substrate to form thin film at room temperature and the substrates include Q235 carbon steel and glass fiber-reinforced epoxy resin (GFE), respectively.

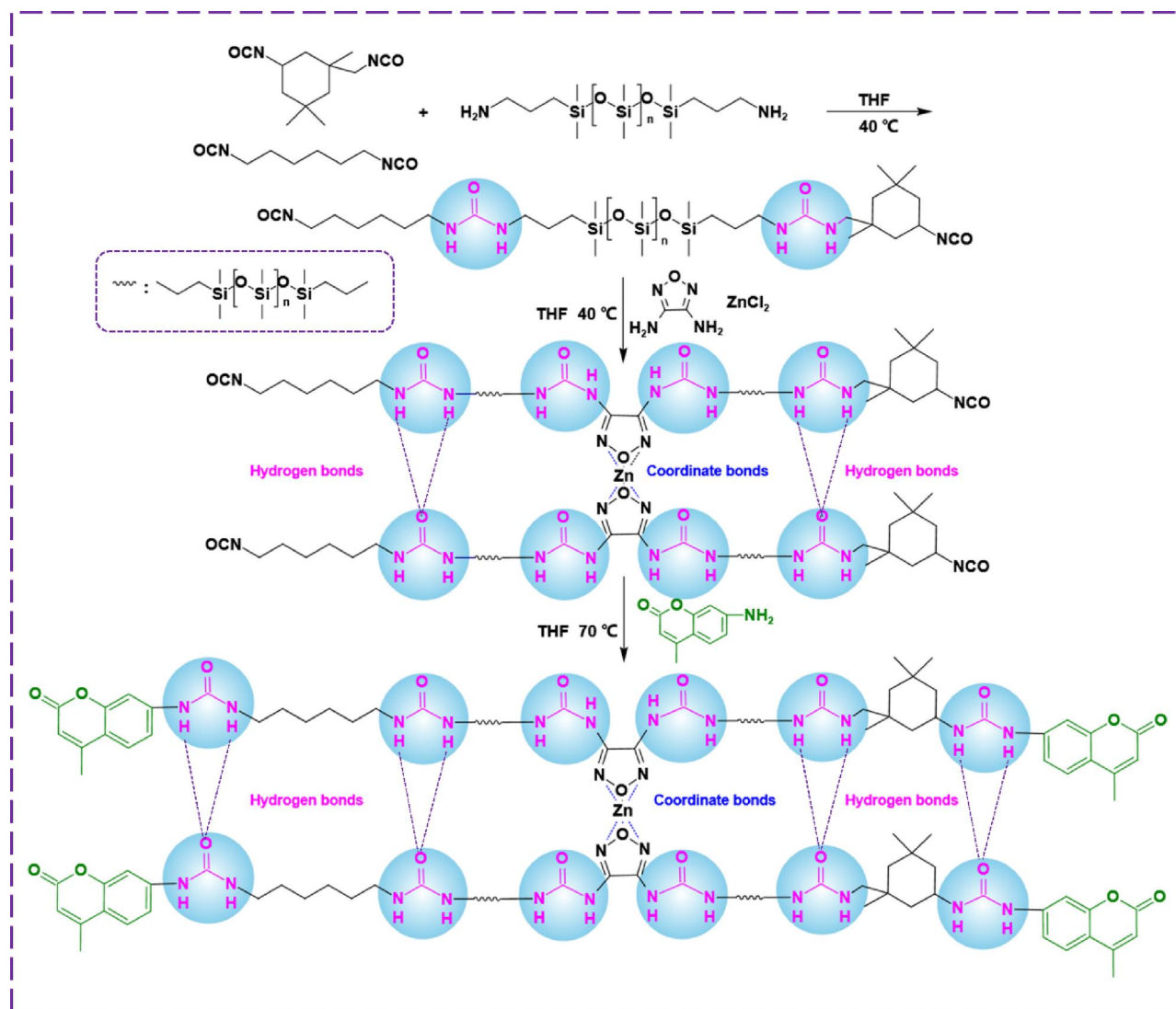


Figure 1. Synthesis routes of the PDMS-PU-AMC-3.

2.4. Characterization

The FT-IR was detected using Nicolet 10 spectrometer under room temperature condition. The samples were scanned and analyzed 32 times within the wavelength range of 4000 cm^{-1} to 400 cm^{-1} . PDMS-PU-AMC-3 was dissolved in CDCl_3 and detected using BRUKER-AVANCE III HD at a frequency of 500 MHz at room temperature. The composition of elements and chemical bonds of the sample were determined using XPS, the calibration spectrum was obtained using the 284.8 eV signal from carbon. The surface morphology of the sample was characterized using SEM (Gemini SEM 500 from Germany). Prior to observation, the sample was sputter-coated with gold. Energy-dispersive X-ray spectroscopy (EDS) maps were recorded on the captured SEM images.

2.5. Mechanical and self-healing properties

The stress-strain behavior and elastic modulus of the PDMS-PU-AMC-x were determined using a universal tensile

testing machine. The samples were cut into a standard dumbbell shape with a mold of $4 \times 75\text{ mm}$. The strain rate was set at 50 mm/min. To evaluate the self-healing properties of the PDMS-PU-AMC-3, the sample was cut into two parts from the middle and then allowed it to self-heal for 6 h at 70°C . The self-healed sample was tested to get its mechanical characteristics. The initial and after self-healing mechanical strength are S_1 and S_2 , respectively. The Self-healing efficiency was calculated by following formula:

$$\text{Self-healing efficiency} = \frac{S_2}{S_1} \times 100\%$$

The surface hardness of the PDMS-PU-AMC-x was measured with a shore hardness tester and sample thickness was approximately 1 mm. The test was repeated for three times for each sample and the average value was finally calculated.

2.6. Swelling ratio

Weighing the initial mass of a square sample ($10 \times 10\text{ mm}^2$) as m_0 , the PDMS-PU-AMC-x coatings were immersed in

artificial seawater for 21 days. The mass of the dried sample is m_1 after removing the residual artificial seawater. The swelling ratios of the PDMS-PU-AMC-x coatings were calculated by the following formula:

$$\text{Swelling ratio} = \frac{m_1 - m_0}{m_0} \times 100\%$$

2.7. Dynamic mechanical property testing and thermo gravimetric analysis

Using DMA850 dynamic mechanical analyzer, the samples were conducted in tension mode at frequency of 1 Hz with heating rate of 3 K/min with the temperature range of 133–353 K.

The TGAQ50 thermal analyzer was employed to estimate the thermo gravimetric properties of the samples. The tests were conducted under nitrogen gas atmosphere with flow rate of 50 mL/min. The heat rate was 10 °C/min and the temperature range is from 0 to 800 °C.

2.8. Water contact angle tests

The self-cleaning test uses Carbon Particles and $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ as contaminants. Contaminants are placed on the surface of PDMS-PU-AMC-3 and removed with a small amount of deionized water. Deionized water and n-hexadecane contact angles (CA) were measured using the German device (Kruss DSA25). The tests were performed at room temperature in 2 μL of deionized water and n-hexadecane with the static sessile drop method. The three CA values (θ) were averaged at different areas for each sample. The surface energy of the coating was calculated by the Owens two-liquid method.

2.9. Adhesion force tests and salt spray tests

According to GB/T5210-1985 standard, a XH-M type pull-off adhesion tester was employed to obtain the adhesion strength for the coatings to the substrate for GFE and Q235 steel, respectively. Aluminum ingots were bonded to the surface of the coating using acrylic adhesive (Cyanoacrylate adhesive, Ergo).

A vertically and uniformly distributed tensile force at a speed of 10 mm/min was exerted to the aluminum ingots of the sample surface to obtain delamination data between the coating and substrate. For each sample, three values from different areas were averaged to determine the adhesion strength.

The salt spray resistance test was assessed by the GB/T1771-2007 standard. The middle part of the carbon steel was coated with PDMS-PU-AMC-3 and the two sides were bare. 3.5 wt% NaCl solution was filled in the salt spray machine at 35 °C and the degree of the carbon steel was observed after 24 h, 72 h, 120 h, 168 h corrosion.

2.10. Anti-fouling tests

Gram-negative bacterium *E. coli* and gram-positive bacterium *S. aureus* were selected as model bacteria.^[32,33] The plate

count method was used to study the antibacterial performance of the coatings against *E. coli* and *S. aureus*. The coverslips (18 × 18 mm²) were named as the control sample. The coverslips coated with PDMS-PU, PDMS-PU-AMC-3 and PDMS-PU-AMC-3(WZn) were placed in a six-well plate about 6 h under ultraviolet light to kill other bacteria. Subsequently, 5 mL *E. coli* and *S. aureus* suspensions (10⁸ CFU/mL) were added to the six-well plate, respectively. The six-well plate was incubated at 37 °C and 60% humidity for 6 h. The coating was washed twice with 10 mL of PBS buffer to remove non-adhesive bacteria. Finally, the coating was subjected to 10 min of ultrasound in 10 mL PBS buffer, then 200 μL bacterial suspension after ultrasound was collected. The elution buffer was evenly dispersed into solid medium and incubated at 37 °C for 24 h. The formation of bacterial colonies was observed to evaluate the antibacterial performances of the coating. The antibacterial efficiency of the coatings is calculated according to the formula:

$$\text{Antibacterial efficiency} = \frac{(N_c - N_s)}{N_c} \times 100\%$$

where N_c and N_s are the number of bacteria on the coverslips without and with coating, respectively.

Microscopy was used to observe the growth and attachment of algae on the surface of the coating, the algae resistance of the coating was evaluated by seawater chlorella. The blank coverslips and coverslips coated with PDMS, PDMS-PU-AMC-3, PDMS-PU-AMC-3(WZn) were placed in a six-well plate with 10 mL chlorella suspension, and incubated in the mode of light for 12 h and dark for 12 h at 25 °C. After 72 h, the unattached algae on the surface of the coating were gently washed off by artificial seawater. The scale of the measurement was 100 μm and the coverage area of algae was counted by image j software.

3. Results and discussion

3.1. FTIR and NMR spectra of the samples

Figure 2a shows the FTIR spectra of the IPDI, HDI, DAF, APT-PDMS and PDMS-PU-AMC-3. The disappearance of the characteristic $\text{N}=\text{C}=\text{O}$ peak at 2243 cm^{-1} confirms the synthesis of silicon-based polyurethane, indicating that the $\text{N}=\text{C}=\text{O}$ group reacted completely with the amino group in the polymer. Furthermore, the peak at 1646 cm^{-1} is related to the stretching vibration of the $\text{C}=\text{N}$ bond, hinting DAF had been successfully entered into the main chain of the PDMS-PU-AMC-3. Compared with the APT-PDMS, the PDMS-PU-AMC-3 exhibited stretching vibration peak of $\text{N}-\text{H}$ at 3340 cm^{-1} , demonstrating the formation of a large number of urea bonds in the polymer. The dense urea bonds bring the excellent mechanical properties of the PDMS-PU-AMC-3. The peak at 1257 cm^{-1} represents the in-plane bending vibration of $\text{Si}-\text{CH}_3$ coming from the APT-PDMS. The peaks at 1100 cm^{-1} are related to asymmetric stretching vibration for the $\text{Si}-\text{O}-\text{Si}$ main chain of the PDMS-PU-AMC-3.^[34]

As shown in Figure S1, the peak at 1646 cm^{-1} for the $\text{C}=\text{N}$ bond of DAF shifted to 1628 cm^{-1} for the

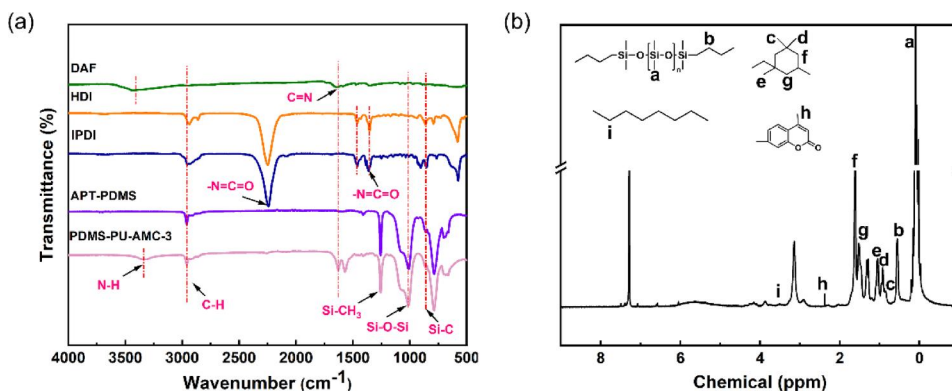


Figure 2. (a) The FT-IR spectra of DAF, HDI, IPDI, APT-PDMS and PDMS-PU-AMC-3. (b) ^1H NMR spectra of the PDMS-PU-AMC-3.

PDMS-PU-AMC-3. The redshift attributes to d electrons on zinc ions feed back to the antibond molecular orbital of $\text{C}=\text{N}$, which reduces bond strength of $\text{C}=\text{N}$. This further confirms the formation of coordination bond between zinc ions and $\text{C}=\text{N}$.

According to the Figure 2b ^1H NMR, there is a sharp and distinct $-\text{Si}(\text{CH}_3)_2-$ characteristic peak at 0.09 ppm, indicating the successful incorporation of organ silicon as soft segment in the PDMS-PU-AMC-3. The specific peaks of $-\text{CH}_3$ and $-\text{CH}_2-$ of IPDI were observed at 0.83, 0.91, 1.05, 1.51 and 1.60 ppm, respectively. The $-\text{CH}_2-$ peak of HDI appears at 3.5 ppm. These results suggest that reactions occur between two isocyanates and amino groups of the APT-PDMS. Furthermore, the peak of $-\text{CH}_3$ of AMC at 2.37 ppm was observed, suggesting AMC has been successfully introduced into the polymer.

3.2. XPS and EDS analysis

The XPS full spectrum as shown in Figure 3a reveals that the PDMS-PU-AMC-3 is composed of elements including C, N, O, Si, and Zn, which aligns with the results obtained from EDS. The C 1s spectrum of the PDMS-PU-AMC-3 has three peaks located at 284.8, 286.5 and 288.5 eV, corresponding to C-C, C-O, and C=O bonds respectively as shown in Figure 3b. The N 1s spectrum can be divided into two peaks, which located at 398.9 and 399.5 eV belonging to N-H and C=N bonds in Figure 3c, respectively. The spectra of C 1s and N 1s hint the PDMS-PU-AMC-3 contain a lot of urea bonds. The O 1s spectrum displays three peaks at 531.1, 531.5 and 532 eV in Figure 3d, corresponding to the C-O, Si-O and C=O bonds.^[35] The Si 2p spectrum in Figure 3e shows two peaks at 101.5 and 101.9 eV concerning the Si-C and Si-O bonds. The Zn 2p spectrum in Figure 3f exhibits peaks at 1021 and 1045 eV, which are assigned to Zn 2p_{3/2} and Zn 2p_{1/2}, respectively. EDS results in Figure 3g explain the uniform distribution of C, N, Zn, Si and O in the PDMS-PU-AMC-3, indicating the formation of the coordination bonds.

3.3. Mechanical properties and self-healing performance

Figure 4a shows the stress-strain curves of the PDMS-PU-AMC- x coatings, demonstrating the influence of different

ratio of IPDI to HDI on the mechanical performances of the PDMS-PU-AMC- x . With the increasing of the ratio of HDI, the tensile strengths of the PDMS-PU-AMC- x decrease and their elongations increase gradually. When the ratio of IPDI to HDI is 1:1, the corresponding PDMS-PU-AMC-3 exhibits ideal comprehensive characteristics which include 1073.1% elongation and 11.8 MPa tensile strength. HDI has longer carbon chain than IPDI which provides better elongation characteristics for the PDMS-PU-AMC-3. However, the PDMS-PU-AMC-5 shows less elongation and a little higher tensile strength compared with the PDMS-PU-AMC-4 which maybe related the regularity of the polymer chains. Interestingly, the value of shore hardness for the PDMS-PU-AMC-3 is the lowest 59.25 A among the five samples as shown in Figure 4b. PDMS-PU-AMC-1 has the highest hardness of 74.5 A, PDMS-PU-AMC-5 has hardness of 71.75 A.^[36] This result suggests the disorder arrangement of HDI and IPDI segments in the long polymer chain as shown in Figure 1, which brings the better mobility for the PDMS-PU-AMC-3 than the other four samples. The active mobility of the polymer chain may generate self-healing effect. From the photos at the same temperature and self-healing time as shown in Figure S2, the scratches remained unchanged for the PDMS because it did not contain dynamic covalent bonds. The regularity of the molecular chains of the PDMS-PU-AMC-1 and PDMS-PU-AMC-5 increased the difficulty of molecular chain movement, resulting in the decrease of self-healing ability. The scratches of the PDMS-PU-AMC-3 disappeared completely after 1 h. This result further confirms the regularity of the PDMS-PU-AMC-3 is the lowest because of the random arrangement of IPDI and HDI segments in the polymer chain. Furthermore, the increased regularity of polymer chains leads to the loss of self-healing capability in PDMS-PU-AMC- x . Figure 4c is the self-healing pictures of the PDMS-PU-AMC-3 at 70 °C for 6 h. The self-healing ability is closely related to chemical bonds between the molecular chains and long chain mobility of the obtained polymer.^[37] Abundant hydrogen bonds between urea bonds and coordinate bonds between DAF segment and zinc ions enhance interactions between molecular chains as shown in Figure 1. At the same time, these dynamic hydrogen and coordination bonds confer the PDMS-PU-AMC-3 good self-healing ability. The self-healing efficiency for the PDMS-PU-AMC-3 can reach 65% of the original

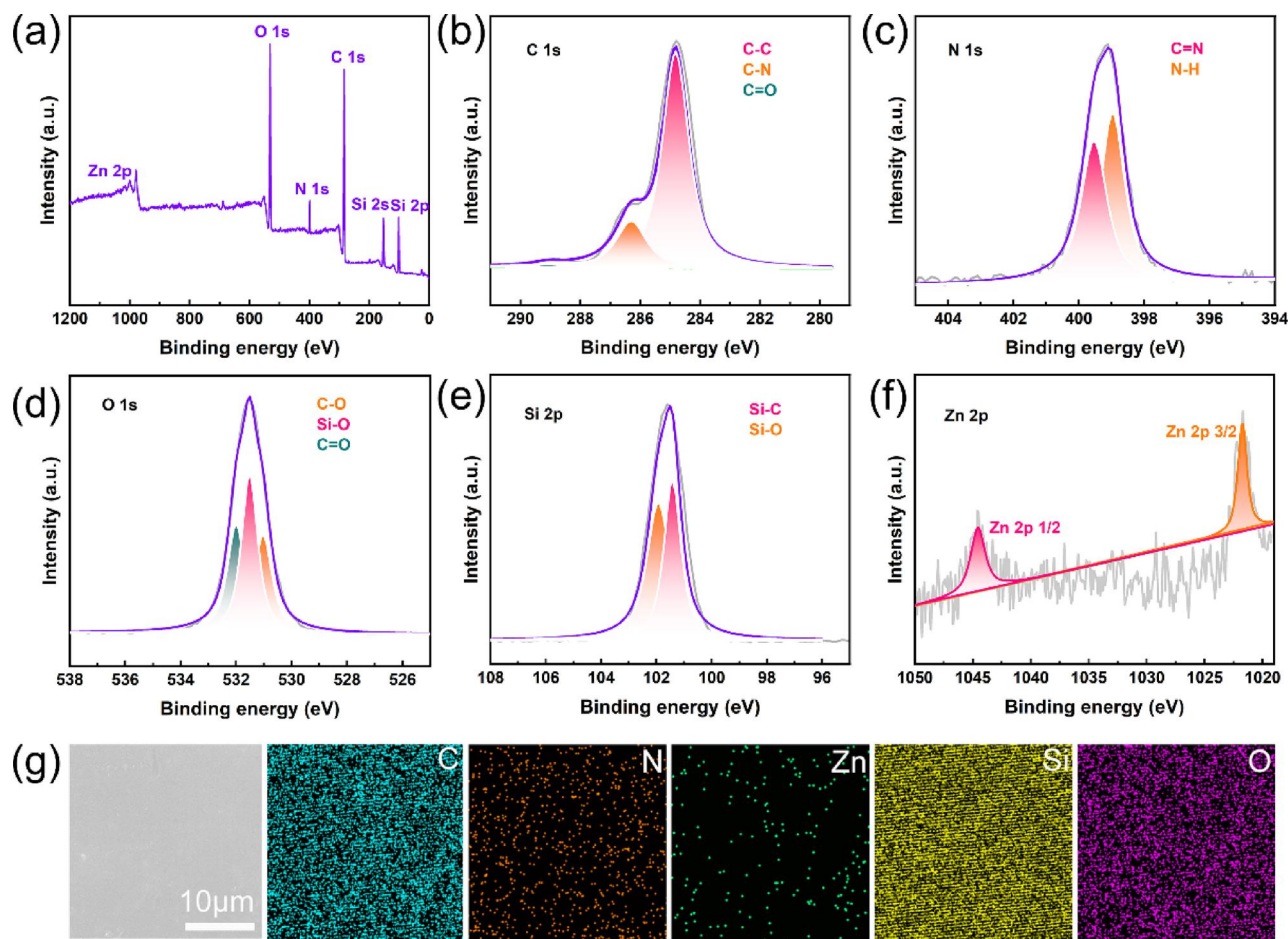


Figure 3. (a) XPS full spectrum of the PDMS-PU-AMC-3. (b–f) High-resolution XPS Spectra of C 1s, N 1s, O 1s, Si 2p and Zn 2p of the PDMS-PU-AMC-3. (g) SEM image and the elemental mapping pictures of the PDMS-PU-AMC-3.

tensile strength as shown Figure 4d. Both excellent mechanical properties and self-healing ability extend the service life of the PDMS-PU-AMC-3 as marine antifouling coating.

Figure 4e exhibits 0.725g PDMS-PU-AMC-3 can easily lift 2.5kg weight and Figure 4f shows the tensile elongation up to 600%. These results further demonstrate the ideal mechanical performances resulted from a large amount of hydrogen and coordination bonds. These interactions between chains are similar to the interlocking of zipper on clothes with mild cross-linking ensuring the mechanical and self-healing characteristics of the obtained polymer. The PDMS-PU-AMC-3 can dissolve in organic solvents such as in THF and absolute ethanol as shown in Figure S3, allowing the recycling use of the coating.

3.4. Thermal stability of PDMS-PU-AMC-x

The high initial storage modulus of the PDMS-PU-AMC-3 indicates that the dense hydrogen bond cross-linking in the polymer system hinders the flow of the polymer chain (Figure 5a), so the obtained PDMS-PU-AMC-3 can be self-healed by heating. The peak temperature of $\tan\delta$ is used to indicate the glass transition temperature (T_g) of the elastomer as shown in Figure 5b. T_g of the PDMS-PU-AMC-3 is -97.57°C . The low T_g provides possible flexibility to the polymer chains, enabling the self-healing for the PDMS-PU-

AMC-3.^[38] The value of $\tan\delta$ is less than 1, indicating that the PDMS-PU-AMC-3 shows mainelasticitydeformation rather than viscosity deformation. The TGA curves in Figure 5c show that the main weight loss begins above 300°C and complete decomposition occurs above 600°C . The DTG curves of the PDMS-PU-AMC-x exhibit two main decomposition peaks as shown in Figure 5d. The first decomposition peak between 300°C and 400°C corresponds to breakage of urea groups and coordination bonds, while the second decomposition between 450 and 600°C is attributed to pyrolysis of the soft segments of the silicone.^[39,40]

3.5. Surface properties

It is widely recognized that the surface properties of materials play a critical role in the attachment of organisms and pollutants in marine pollution. The material's static anti-fouling ability is evaluated through self-cleaning test. Figure 6a demonstrates that carbon articles and $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ easily slide off glass slide coated with the PDMS-PU-AMC-3. There are no residues on the surface of the coating. The self-cleaning test shows that the PDMS-PU-AMC-3 has good static anti-fouling ability. Figure 6b is the surface energy of the five samples and Table S2 gives the values which are in the range of 20–25 mN/m. Figure 6c and Figure S4 exhibit water contact angles of PDMS-PU-AMC-x

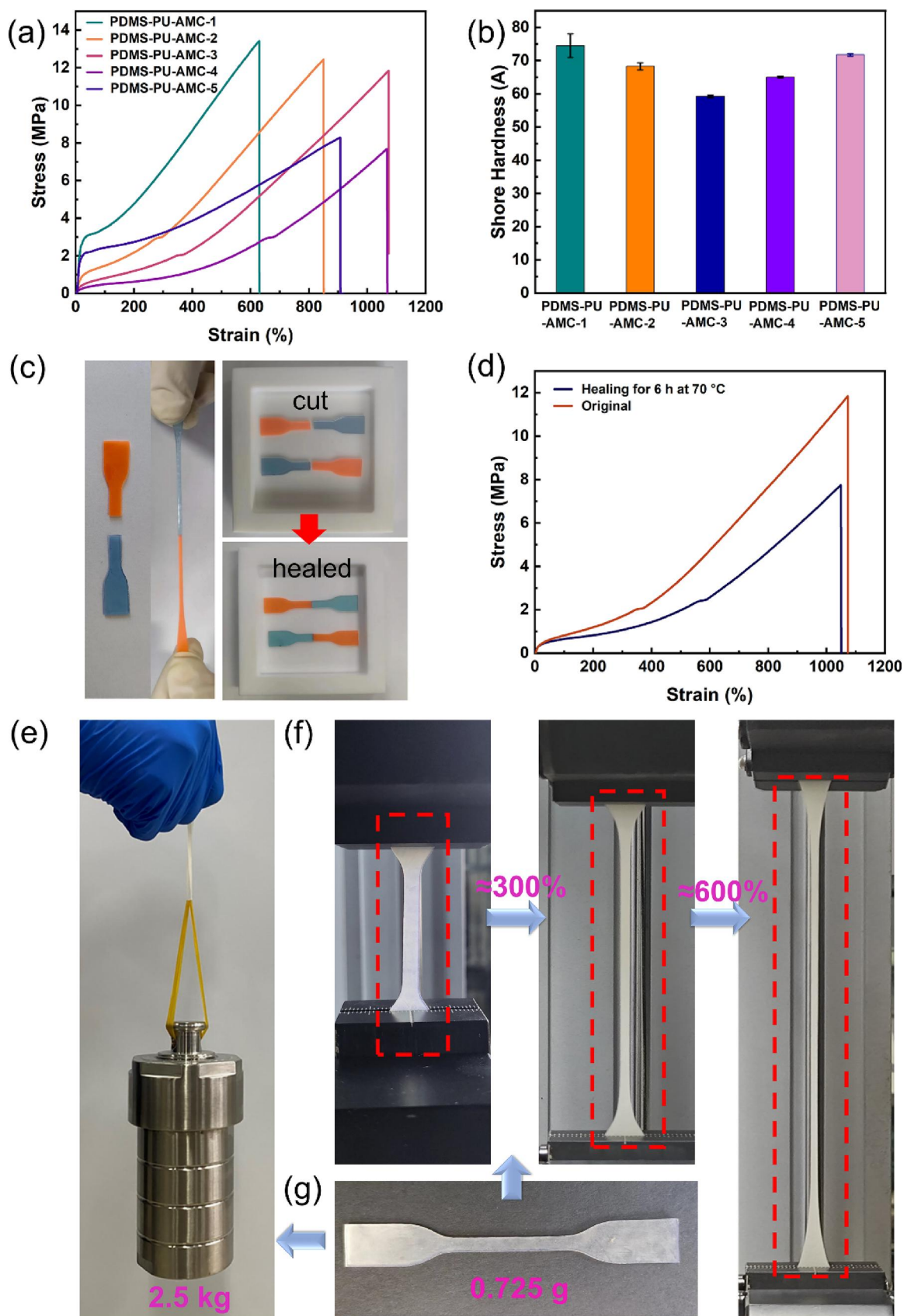


Figure 4. The mechanical properties, shore hardness and self-healing performances of the PDMS-PU-AMC-x. (a) Stress-strain curves of the PDMS-PU-AMC-x at room temperature. (b) Shore hardness of the PDMS-PU-AMC-x. (c) Photographs of the PDMS-PU-AMC-3 before and after self-healing. (d) The stress-strain curves before and after self-healing of the PDMS-PU-AMC-3. (e) Image of the PDMS-PU-AMC-3 lifting a heavy object. (f) Stretched pictures at different tensile ratios for the PDMS-PU-AMC-3. (g) 0.725 g dumb-bell shaped PDMS-PU-AMC-3.

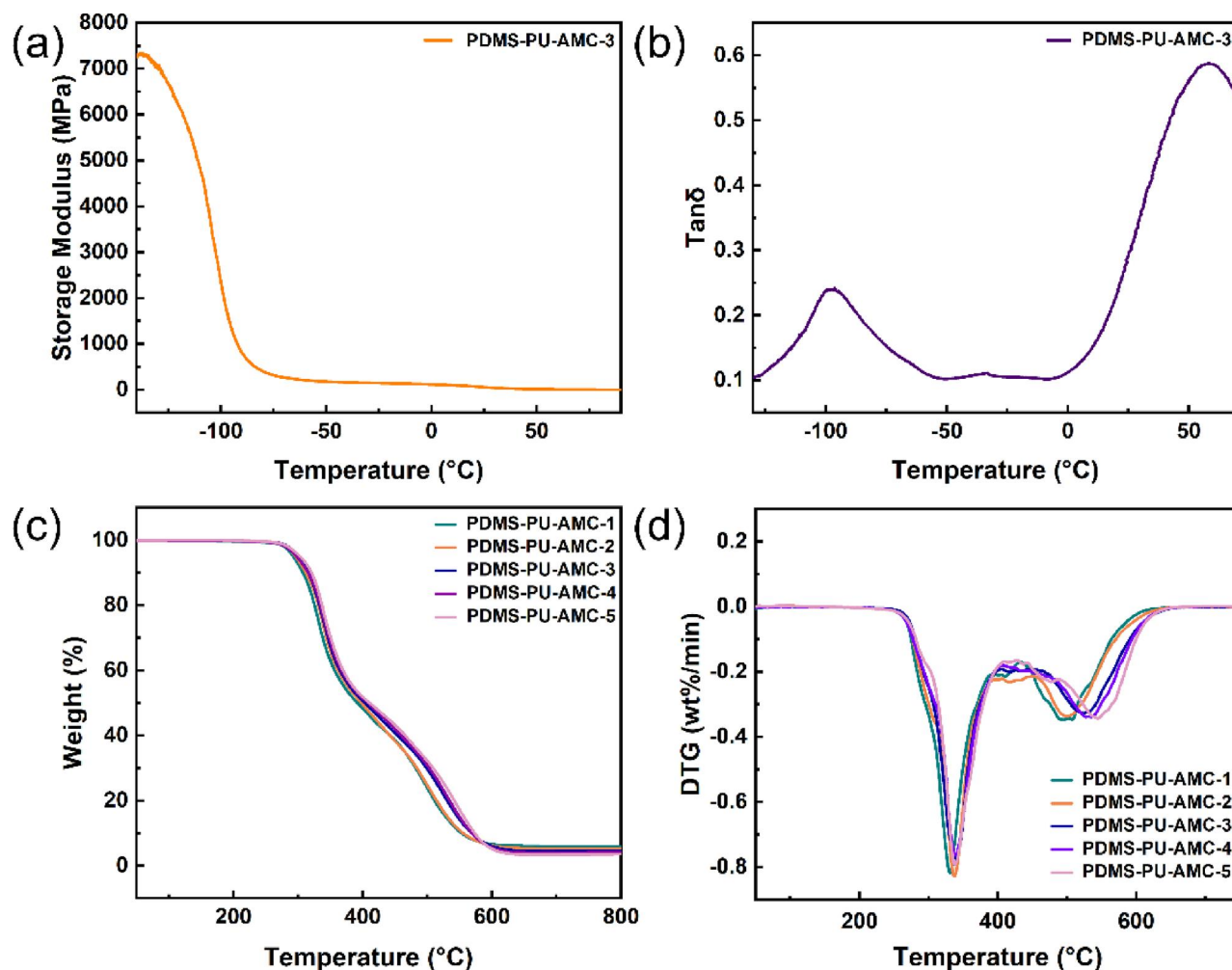


Figure 5. (a) E' -temperature curve of the PDMS-PU-AMC-3. (b) $\tan \delta$ -temperature curve of the PDMS-PU-AMC-3. (c) TGA curves of the PDMS-PU-AMC-x. (d) DTG curves of the PDMS-PU-AMC-x.

coatings which are around 100° as listed in Table S2. The PDMS-PU-AMC-3 has the lowest surface energy of only 20.76 mN/m which is beneficial to improve the antifouling effect as marine coating.^[41]

The surface roughness of the coating affects the adhesion of fouling organisms. The surface with low roughness can effectively reduce the adhesion of marine fouling organisms. Figure S5 displays 2D, 3D images and R_a values of the PDMS-PU-AMC-3 before and after swelling in 3.5 wt% NaCl artificial seawater for seven days, the R_a value of the PDMS-PU-AMC-3 is 0.3 nm before swelling, the R_a value of the PDMS-PU-AMC-3 increases slightly to 2.07 nm after swelling. Figure 6d presents the swelling ratios of PDMS-PU-AMC-x coatings after being immersed in 3.5 wt% NaCl artificial seawater about 21 days. The results show a slight weight increase of the PDMS-PU-AMC-x coating. The swelling rate of the PDMS-PU-AMC-x coating was under 2%, indicating that the coating has good seawater resistance.

3.6. Salt spray resistance of PDMS-PU-AMC-3 coating

The actual corrosion resistance of PDMS-PU-AMC-3 anti-fouling coatings can be validated through salt spray test.

The PDMS-PU-AMC-3 is coated on the middle of the Q235 carbon steel and the two sides of the Q235 carbon steel are bare. 3.5 wt% NaCl artificial seawater was filled in the machine which mimics the conditions in real marine environment. In Figure 7a, it is evident that the coated portion with the PDMS-PU-AMC-3 has no obvious corrosion after 24, 72, 120 and 168h. Furthermore, the coating didn't peel off illustrating good adhesion strength between the PDMS-PU-AMC-3 and the substrate. In comparison, the areas without the PDMS-PU-AMC-3 coating experienced severe corrosion, further telling that the PDMS-PU-AMC-3 coating is highly suitable for marine anti-fouling and anti-corrosion applications.

3.7. Adhesion properties

High adhesion was another key factor for polymer coatings to ensure their integrity and durability.^[42,43] Figure 7b and Table S3 shows the adhesion strength of PDMS-PU-AMC-x coatings on carbon steel and GFE. The PDMS-PU-AMC-3 coating exhibited superior adhesion to carbon steel is 2.721 ± 0.130 MPa. The surface of the ship is usually coated with epoxy coating, and the strong adhesion for the

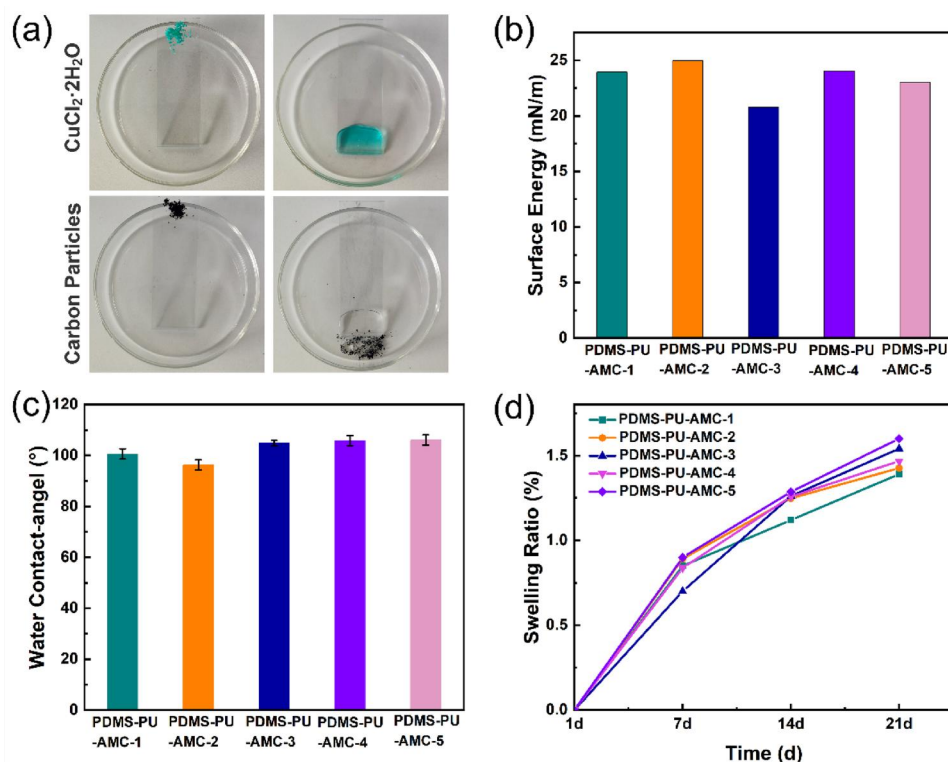


Figure 6. Surface properties of the PDMS-PU-AMC-3. (a) Photographs of carbon articles and $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ sliding off slides coated with the PDMS-PU-AMC-3. (b) Surface energy values of the PDMS-PU-AMC-x coatings. (c) Water contact angles of the PDMS-PU-AMC-x coatings. (d) The swelling ratio curves of the PDMS-PU-AMC-x coatings soaked in artificial seawater for 21 days.

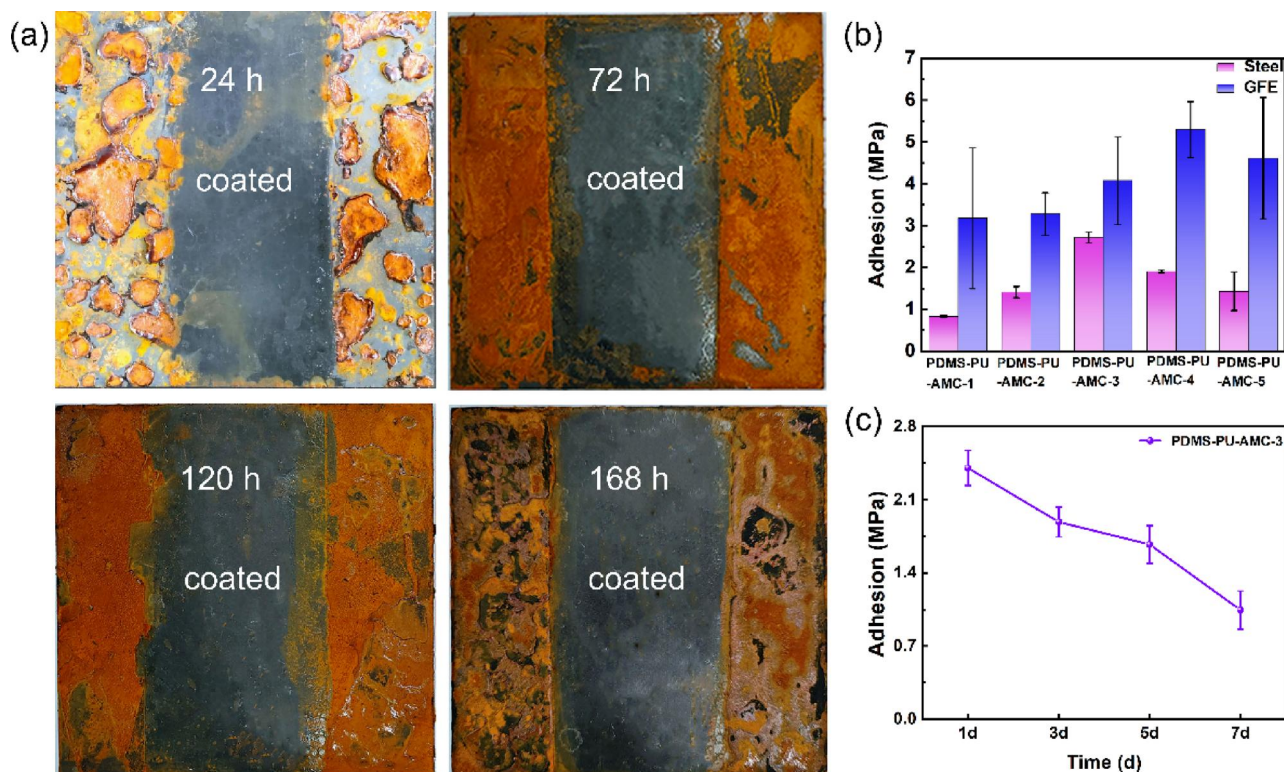


Figure 7. (a) Degree of corrosion of carbon steel coated with PDMS-PU-AMC-3 in 24 h, 72 h, 120 h, 168 h. (b) Adhesion of all coatings in carbon steel and GFE. (c) Adhesion on carbon steel coated with PDMS-PU-AMC-3 at different corrosion times.

PDMS-PU-AMC-3 coatings on the GFE is about 4.078 ± 1.050 MPa. High adhesion is attributed to the abundance of urea bonds, which enables strong interactions with the

substrate through hydrogen bonds.^[44] The adhesion strength of the PDMS-PU-AMC-x coatings on carbon steel surpass 0.8 MPa, and remains above 3.0 MPa on GFE as shown in

Table S3. From Figure 7b, the PDMS-PU-AMC-3 exhibits the best comprehensive adhesion performances on carbon steel and GFE, meeting the required standards for marine anti-fouling coatings. However, with the increase of corrosion time, the adhesion of the PDMS-PU-AMC-3 coating gradually decreased from 2.416 ± 0.174 MPa in 1d to 1.131 ± 0.193 MPa in 7d from Figure 7c. Addressing the problem of reduced adhesion of the coating in prolonged salt spray corrosive environments remains challenging.

3.8. Anti-fouling properties

Bacteria were the most common marine fouling organisms. The antibacterial ability of the coating was crucial to prevent marine biofouling.^[45,46] The antibacterial efficiency of the PDMS-PU, PDMS-PU-AMC-3, and PDMS-PU-AMC-3 (W Zn) coatings were evaluated against *E. coli* and *S. aureus*. The blank coverslip was named as the control sample. As shown in Figure 8a and b, the antibacterial efficiency of the PDMS-PU is approximately 80% and 96% against *S. aureus* and *E. coli*, respectively. These values are up to 99.5% and 96.7% for the PDMS-PU-AMC-3 because of synergistic effect between hybrid ring of DAF and benzopyranic ring of AMC.^[47,48] The antibacterial efficiency of the PDMS-PU-AMC-3 (W Zn) coating decreased slightly to 99% and 95.5%. Zinc ions can disrupt the metabolism of cell membranes, leading to bacteria to lose their biological functions until bacteria death.^[49,50] Even if without AMC and ZnCl_2 ,

the number of the bacteria on the PDMS-PU coating would also reduce due to the low surface energy.

The excellent antibacterial characteristics of the PDMS-PU-AMC-3 can be explained from the following two main aspects. The low surface energy makes it difficult for the bacteria absorb on the coating surface with the external shear force. The synergistic effect of the zinc ions, AMC, and DAF in the PDMS-PU-AMC-3 inhibits significantly the bacteria attachment on the coating. The ideal antibacterial effect reduces the aggregation of the marine organisms and prolongs the service term of the ships.

The anti-algae properties of the coating were estimated by using chlorella and the microscopic images display lots of chlorellas adhering to the surface of the material as shown in Figure 8c. Figure 8d gives the coverage area of algae of the Control, PDMS, PDMS-PU-AMC-3 and PDMS-PU-AMC-3 (W Zn) are $7.42 \pm 5.18\%$, $13.13 \pm 5.89\%$, $1.23 \pm 0.34\%$, $2.86 \pm 0.45\%$, respectively. Due to the very smooth surface of the blank glass, so algae are not easy to adhere to the surface. However, the static antifouling effect of the PDMS is not ideal, the algae coverage rate of the blank coverslips is lower than that of the coverslips coated with PDMS. It can be clearly seen that the PDMS-PU-AMC-3 coating has the lowest algae coverage rate for only 1.23%, hinting the multi-functional antifouling strategy may achieve an ideal antifouling effect. The final results show that the PDMS-PU-AMC-3 coating has the potential to be applied in marine antifouling field.

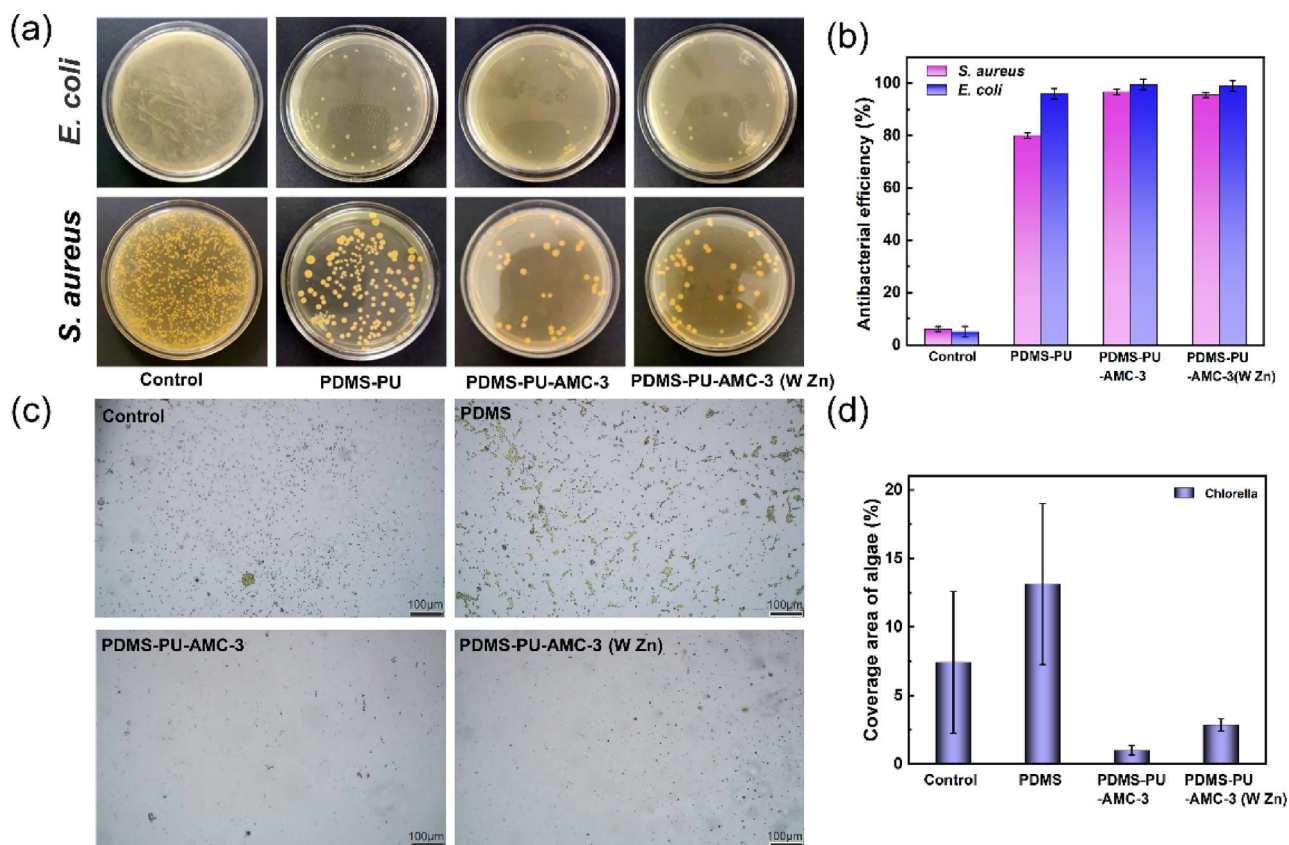


Figure 8. (a) Pictures of *E. coli* and *S. aureus* growing on the Control, PDMS-PU, PDMS-PU-AMC-3 and PDMS-PU-AMC-3 (W Zn). (b) Antibacterial efficiency of the Control, PDMS-PU, PDMS-PU-AMC-3 and PDMS-PU-AMC-3 (W Zn). (c) Microscopic images of algae adhesion for different coatings. (d) The coverage area of algae of the Control, PDMS, PDMS-PU-AMC-3 and PDMS-PU-AMC-3 (W Zn).

4. Conclusion

A novel high mechanical strength organic-silicone polyurethane marine antifouling coating has been prepared with facile two steps. Two different isocyanates act as hard segments to provide the mechanical strength of the material, while the organic silicone chains in the PDMS-PU-AMC-3 work as soft segments to impart flexibility. The abundant urethane bonds formed by the reaction between the amino groups in DAF and APT-PDMS with isocyanates enhance the intermolecular interactions and improve the adhesion strength. The mechanical robustness, self-healing ability and antifouling capacity achieve the optimal state when the ratio of IPDI to HDI is 1:1. This silicone-based coating exhibits excellent corrosion resistance, antifouling and antibacterial properties under static conditions mainly brought by DAF and AMC. The synergistic effect between DAF and AMC endows excellent and long-term antibacterial and antifouling properties. This well-designed coating provides great potential in biomedical and marine antifouling application.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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